

Lecture 5

Action-angle variables in practice

Objectives. *Lecture 4 proved that the motion of an integrable system lives on tori. This lecture computes on them. We define the action variables, prove that they are well defined and that their derivatives are the periods, and build the conjugate angles from a generating function. Then the oscillator and the pendulum, in both of its regimes, done completely; and for Kepler the radial action and the frequencies — which all coincide, and that is why its orbits close.*

5.1 Action variables, and why they are well defined

Imported: from [Lecture 4](#), the $\mathbb{R}^n$ action Φ_t on a torus $\Sigma \cong \mathbb{R}^n/\Lambda$, with $\partial\Phi_t(x_0)/\partial t_i = X_{F_i}(\Phi_t(x_0))$ at every t , a basis e_i of the period lattice, the cycles $\gamma_i(s) = \Phi_{se_i}(x_0)$ and $A = (e_1 \cdots e_n)$ ([Theorem 4.8](#), [Definition 4.13](#)); and $p \cdot dq$ closed on Σ ([Proposition 4.14](#)). From [Lecture 3](#), $d(p \cdot dq) = -\omega$ ([Theorem 3.28](#)), $\omega(X_f, v) = df(v)$ ([Definition 3.5](#)), and type-2 generating functions ([Proposition 3.16](#), (3.11)). From [Lecture 2](#), deformation invariance ([Corollary 2.26](#)).

We work near one such torus Σ , at the value f . The phase space is $\mathbb{R}^{2n}$ or a cotangent bundle, as [Proposition 4.14](#) requires; all three examples below are. One fact about the neighbourhood is imported.

Standing assumption (cited: Arnold, §50 A, the problem there). For f' near f , the component of $\mathcal{M}_{f'}$ near Σ is again a torus satisfying the hypotheses of [Theorem 4.8](#), and the base point x_0 and the lattice basis e_i can be chosen to depend smoothly on f' . The smooth lattice basis is the implicit function theorem applied to $\Phi_t(x_0) = x_0$; that the nearby components are tori is the content of Arnold's problem. We write out neither.

Definition 5.1 (Action variables). The action variables of the torus at f are

$$I_i(f) = \frac{1}{2\pi} \oint_{\gamma_i} p \cdot dq, \quad i = 1, \dots, n, \quad (5.1)$$

each cycle traversed in the direction of increasing s .

This is the integral (2.1) that [Lecture 2](#) promised, and [Lecture 2](#) settled the first two of the three things it needs. The next proposition supplies the third.

Proposition 5.2 (The actions are well defined · proved modulo Stokes).

1. I_i does not change if γ_i is replaced by any loop into which it can be deformed within Σ . In particular it does not depend on the base point x_0 .
2. I_i depends only on f : it is a function of $F_1, \dots, F_n$, and so a first integral. The I_i are in involution.

Proof. (1) is Proposition 4.14 together with Corollary 2.26, which rests on Stokes. For the base point: if $x_1 = \Phi_\tau(x_0)$, the loops $s \mapsto \Phi_{u\tau}(\Phi_{se_i}(x_0))$, $u \in [0, 1]$, deform the cycle based at x_0 into the cycle based at x_1 , inside Σ . (2) The definition uses only the torus at f , so $I_i = I_i(F)$. For two functions of F , Leibniz and the chain rule give $\{I_i, I_j\} = \sum_{a,b} \frac{\partial I_i}{\partial f_a} \frac{\partial I_j}{\partial f_b} \{F_a, F_b\} = 0$. ■

A different basis of Λ gives different cycles and different actions: the new basis is $e'_i = \sum_j U_{ij} e_j$ with U an integer matrix of determinant ± 1 , and $I' = UI$. There is no preferred choice in general. What never changes is what the actions are *for*: they are functions of F whose derivatives are the periods.

Proposition 5.3 (The period formula). *With A the matrix whose columns are the basis vectors $e_i(f)$,*

$$2\pi \frac{\partial I_i}{\partial f_j} = (e_i)_j, \quad \text{that is,} \quad \frac{\partial I}{\partial f} = \frac{1}{2\pi} A^\top. \quad (5.2)$$

Proof. Fix i and j , and sweep the cycle as f moves in direction j : $x(s, u) = \Phi_{se_i(u)}(x_0(u))$, with $x_0(u)$ the base point on the torus at $f + u \hat{e}_j$ ($\hat{e}_j$ the j -th standard basis vector of $\mathbb{R}^n$) and $e_i(u) = e_i(f + u \hat{e}_j)$. Pull $p \cdot dq$ back to the (s, u) -rectangle, where it reads $a ds + b du$ for two functions a, b . Then $2\pi I_i(f + u \hat{e}_j) = \int_0^1 a ds$, and

$$2\pi \frac{dI_i}{du} = \int_0^1 \partial_u a ds = \int_0^1 (\partial_u a - \partial_s b) ds,$$

since $\int_0^1 \partial_s b ds = b(1, u) - b(0, u) = 0$: each loop closes, $x(1, u) = x(0, u)$. The integrand is the pullback of $d(p \cdot dq) = -\omega$ evaluated on (∂_u, ∂_s) , by (2.10) and because d commutes with pullback (2.15). So it equals $-\omega(\partial_u x, \partial_s x) = \omega(\partial_s x, \partial_u x)$. Now $\partial_s x = \sum_k (e_i)_k X_{F_k}$, and the flows preserve F , so $F_k(x(s, u)) = f_k + u \delta_{kj}$. Hence

$$\omega(\partial_s x, \partial_u x) = \sum_k (e_i)_k dF_k(\partial_u x) = \sum_k (e_i)_k \partial_u (f_k + u \delta_{kj}) = (e_i)_j,$$

constant in s . Integrating over $s \in [0, 1]$ gives (5.2). ■

Corollary 5.4 (Frequencies are derivatives of the energy). *The I_i are independent near Σ , so f , and with it $H = h(f)$, is a function of the actions: $H = K(I)$. The frequency vector of (4.9) is*

$$\nu(I) = \frac{\partial K}{\partial I}. \quad (5.3)$$

With one degree of freedom, $dI/dE = T/2\pi$ and $\nu = dE/dI = 2\pi/T$.

Proof. $\det(\partial I/\partial f) = \det A/(2\pi)^n \neq 0$, since the e_i are a basis; the inverse function theorem makes f a smooth function of I , with $\partial f/\partial I = 2\pi(A^\top)^{-1}$. By the chain rule $\partial K/\partial I_i = \sum_j \partial_j h \partial f_j/\partial I_i$, that is, $\partial K/\partial I = (2\pi(A^\top)^{-1})^\top \nabla h = 2\pi A^{-1} \nabla h$, which is (4.9). For $n = 1$, $F = H$ and the lattice is $T\mathbb{Z}$, so $A = (T)$. ■

The angles, from a generating function

Theorem 5.5 (Action–angle coordinates · sketched). *Near Σ there are functions $\varphi_1, \dots, \varphi_n$ defined modulo 2π , such that (φ, I) are canonical coordinates, $\omega = \sum_i d\varphi_i \wedge dI_i$. In them $H = K(I)$ and*

$$\dot{I}_i = 0, \quad \dot{\varphi}_i = \nu_i(I) = \frac{\partial K}{\partial I_i}. \quad (5.4)$$

Sketch. Near a point of Σ where the torus is a graph over q , write it as $p = \sigma(q; I)$. It is Lagrangian (Theorem 4.8), so σ is closed (Proposition 3.25) and locally $\sigma = d_q S$ with

$$S(q, I) = \int_{q_0}^q p \cdot dq \quad (\text{along the torus with actions } I). \quad (5.5)$$

Take S as a type-2 generating function with new momenta $P = I$. By Proposition 3.16 the new positions $\varphi_i = \partial S / \partial I_i$ make (φ, I) canonical, and since the graph lies in the level set, $H(q, \partial S / \partial q) = K(I)$: this S solves Hamilton–Jacobi (3.11). Hamilton’s equations for $K(I)$ are (5.4).

S is not a function on the torus. Continued once round γ_j , it gains $\oint_{\gamma_j} p \cdot dq = 2\pi I_j$, by (5.1). So $\varphi_i = \partial S / \partial I_i$ gains $2\pi \partial I_j / \partial I_i = 2\pi \delta_{ij}$: the φ_i are angles, defined modulo 2π , and round γ_j only φ_j turns. Where the torus is not a graph over q — at turning points — one uses a mixed set of coordinates instead; that routine patching is what is omitted (Arnold, §50 C). ■

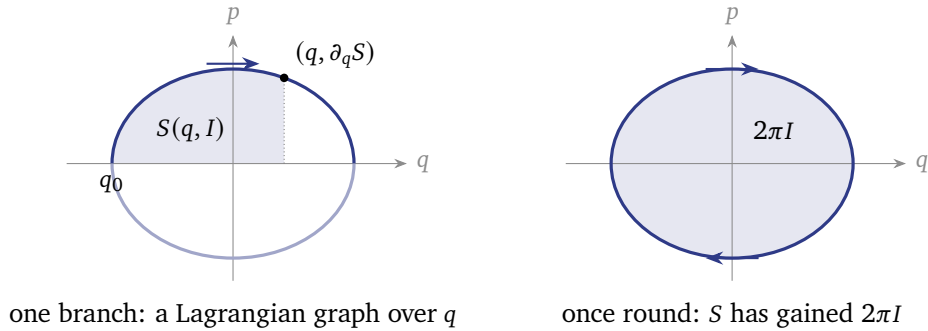


Figure 5.1: The generating function $S(q, I)$ of (5.5), for a torus that is a curve. Left: over the upper branch the curve is the graph of $p = \partial_q S$, and S is the shaded area. Right: once round the cycle, clockwise, S has gained the enclosed area $2\pi I$ — which is why $\varphi = \partial S / \partial I$ is an angle.

The practical route: separation of variables

In practice nobody computes the lattice first. If the Hamilton–Jacobi equation *separates* in some coordinates $q_1, \dots, q_n$ — if $S = \sum_i S_i(q_i)$, so that on the level set each p_i is a function of its own q_i and of the constants f only — then the torus is a product of closed curves C_i in the separate (q_i, p_i) -planes. Going once round C_i with the other coordinates frozen is a cycle on the torus, and the actions are one-dimensional integrals,

$$I_i = \frac{1}{2\pi} \oint_{C_i} p_i dq_i. \quad (5.6)$$

The recipe for the rest of the lecture is: *choose the cycle; integrate; invert to get $K(I)$; differentiate to get ν .*

5.2 The harmonic oscillator

$H = \frac{1}{2}(p^2 + \nu^2 q^2)$, the oscillator of Lecture 1. The level set at $E > 0$ is an ellipse with semi-axes $\sqrt{2E}/\nu$ and $\sqrt{2E}$, traversed clockwise. By Proposition 5.2 and the sign count of

§ 3.7 the action along the motion is the enclosed area over 2π (Figure 5.2):

$$I = \frac{1}{2\pi} \pi \frac{\sqrt{2E}}{\nu} \sqrt{2E} = \frac{E}{\nu}, \quad K(I) = \nu I, \quad \frac{dK}{dI} = \nu.$$

The angle: on the upper branch $S = \int^q \sqrt{2\nu I - \nu^2 q^2} dq$, so $\varphi = \partial S / \partial I = \arcsin(q\sqrt{\nu/2I})$, that is,

$$q = \sqrt{2I/\nu} \sin \varphi, \quad p = \sqrt{2\nu I} \cos \varphi.$$

The Jacobian of $(\varphi, I) \mapsto (q, p)$ is $\cos^2 \varphi + \sin^2 \varphi = 1$, so $dq \wedge dp = d\varphi \wedge dI$: canonical, as Theorem 5.5 promised. This is the reference case: the frequency does not depend on the action. Lecture 1 observed that the oscillator's period is the same on every ellipse; now we see that is the statement $K''(I) = 0$, and the pendulum breaks it.

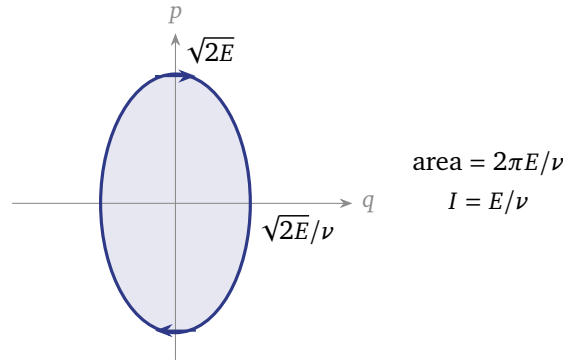


Figure 5.2: The oscillator's action is the area of its orbit over 2π . The orbit runs clockwise, which makes it the negatively oriented boundary; with $d(p \cdot dq) = -\omega$ the two signs cancel and $I > 0$.

5.3 The pendulum, both regimes

$H = \frac{1}{2}p^2 - \cos \theta$ on the cylinder $S^1 \times \mathbb{R}$ ((1.2), (1.1)). With one degree of freedom every regular compact level-set component is a circle and the orbit itself is the cycle, so there is no choice of basis to make. The difficulty is different: *the cycle is not the same kind of loop at every energy* (Figure 5.3).

- $-1 < E < 1$, *libration*: one closed curve round $(0, 0)$. It bounds a disc on the cylinder, and $2\pi I$ is the area it encloses.
- $E > 1$, *rotation*: two components, $p > 0$ and $p < 0$, each going once round the cylinder. Neither bounds anything. Deforming the curve $p = \sqrt{2(E + \cos \theta)}$ down to the zero section $p = 0$, on which $p d\theta$ vanishes, sweeps out the strip beneath it, and $2\pi I$ is the area of that strip — the difference of the two integrals, by Stokes on the strip, both loops taken in the direction of motion. The branch $p < 0$ is its mirror image and has the same action: it runs the other way round the cylinder with $p < 0$, so $p d\theta > 0$ again.
- $E = 1$ is the separatrix, where the level set is not a circle, and $E = -1$ the equilibrium.

Students reliably merge the two regimes into one formula. They cannot be merged: they integrate over loops of different kinds, and the result jumps.

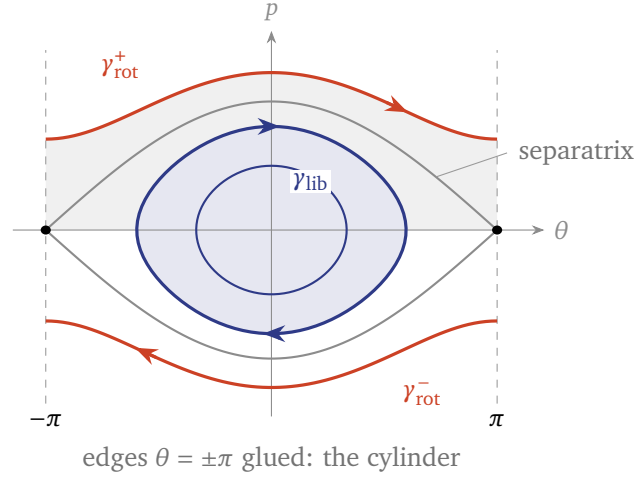


Figure 5.3: The pendulum's two kinds of cycle, on the cylinder unrolled ($\theta = \pm\pi$ are the same line, and the dot there is one point, the unstable equilibrium). Libration curves (indigo, $E = -0.5$ and $E = 0.3$) bound discs, and the action is the enclosed area over 2π . Rotation curves (red, $E = 2$) go round the cylinder and bound nothing; the action is the area of the grey strip between the curve and $p = 0$, over 2π . Arrows: the direction of motion. Libration curves are drawn from (5.10).

The complete elliptic integrals

Both regimes lead to the same two integrals, and they are *defined* here, not assumed.

Definition 5.6 (Complete elliptic integrals). For $0 \leq k < 1$, the complete elliptic integrals of the first and second kind, in the *modulus* k , are

$$K(k) = \int_0^{\pi/2} \frac{d\psi}{\sqrt{1 - k^2 \sin^2 \psi}}, \quad E(k) = \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 \psi} d\psi. \quad (5.7)$$

Upright $E(k)$, always printed with its argument, is the elliptic integral; italic E is the energy. A *trap*: scipy's `ellipk` and mpmath's `ellipk` take the *parameter* $m = k^2$, not the modulus. What we need about them: $K(0) = E(0) = \pi/2$; $E(1) = 1$; and as $k \rightarrow 1$, with $k' = \sqrt{1 - k^2}$,

$$K(k) = \ln \frac{4}{k'} + o(1) \quad (5.8)$$

(cited: DLMF 19.12.1). K diverges because at $k = 1$ the integrand becomes $1/\cos \psi$.

Libration

Proposition 5.7 (The librating pendulum). For $-1 < E < 1$ put $k^2 = (1 + E)/2$. Then

$$T = 4K(k), \quad I = \frac{8}{\pi} \left[E(k) - (1 - k^2) K(k) \right]. \quad (5.9)$$

Proof. On the level set, $E + \cos \theta = E + 1 - 2 \sin^2(\theta/2) = 2(k^2 - \sin^2(\theta/2))$, so $p = 2\sqrt{k^2 - \sin^2(\theta/2)}$ on the upper branch, and the turning points are $\sin(\theta_0/2) = \pm k$. Substitute

$$\sin(\theta/2) = k \sin \psi, \quad \psi \in [-\pi/2, \pi/2]. \quad (5.10)$$

Then $p = 2k \cos \psi$ and $d\theta = 2k \cos \psi d\psi / \sqrt{1 - k^2 \sin^2 \psi}$, so $d\theta/p = d\psi / \sqrt{1 - k^2 \sin^2 \psi}$. The orbit is four copies of the quarter from $\theta = 0$ to θ_0 , and $T = \oint d\theta/\dot{\theta} = \oint d\theta/p$, so $T = 4K(k)$. For the action, the area is four times the area under the upper branch from 0 to θ_0 :

$$I = \frac{4}{2\pi} \int_0^{\theta_0} p d\theta = \frac{2}{\pi} \int_0^{\pi/2} \frac{4k^2 \cos^2 \psi}{\sqrt{1 - k^2 \sin^2 \psi}} d\psi.$$

Write $k^2 \cos^2 \psi = (1 - k^2 \sin^2 \psi) - (1 - k^2)$; the integral splits into $E(k) - (1 - k^2)K(k)$. ■

Check: small amplitude. As $E \rightarrow -1$, $k \rightarrow 0$ and $T \rightarrow 4 \cdot \pi/2 = 2\pi$: the linearised pendulum is the oscillator with $\nu = 1$. Expanding (5.7) to first order in k^2 gives $E(k) - (1 - k^2)K(k) = \frac{\pi}{4}k^2 + O(k^4)$, so $I \approx 2k^2 = E + 1$: the oscillator's $I = E/\nu$, with the energy measured from the bottom of the well. But now T grows with the amplitude, since $K(k)$ increases with k ; the pendulum's frequency depends on its action.

Rotation

Proposition 5.8 (The rotating pendulum). For $E > 1$ put $k^2 = 2/(1 + E)$, again a modulus below 1. Then, on each of the two components,

$$T = 2k K(k), \quad I = \frac{4}{\pi k} E(k), \quad (5.11)$$

where T is the time for θ to advance by 2π .

Proof. Now

$$E + \cos \theta = (E + 1)(1 - k^2 \sin^2(\theta/2)),$$

so on the branch $p > 0$, $p = (2/k)\sqrt{1 - k^2 \sin^2(\theta/2)}$. Put $\psi = \theta/2$, which runs over $[0, \pi]$ as θ runs round once, and use the symmetry of the integrands about $\psi = \pi/2$:

$$\begin{aligned} T &= \int_0^{2\pi} \frac{d\theta}{p} = k \int_0^\pi \frac{d\psi}{\sqrt{1 - k^2 \sin^2 \psi}} = 2k K(k), \\ I &= \frac{1}{2\pi} \int_0^{2\pi} p d\theta = \frac{2}{\pi k} \int_0^\pi \sqrt{1 - k^2 \sin^2 \psi} d\psi = \frac{4}{\pi k} E(k). \end{aligned} \quad \blacksquare$$

Tutorial

The rotating pendulum at high energy

As $E \rightarrow \infty$, $k \rightarrow 0$ and $I \approx \sqrt{2E}$, $T \approx 2\pi/\sqrt{2E}$: a free rotor with $p = \sqrt{2E}$, which does not feel gravity. In both regimes the reader can differentiate (5.9) and (5.11) and find $dI/dE = T/2\pi$, as Corollary 5.4 says it must.¹

The separatrix: where the torus degenerates

At $E = 1$ the level set is the separatrix $p = \pm 2 \cos(\theta/2)$: the unstable equilibrium $(\pi, 0)$, where $dH = 0$, and two orbits that leave it and return to it. It is not a circle, and Theorem 4.8

¹scripts/check-l5-pendulum.py, check 2 — a numerical check at four energies, not a proof; the proof is Proposition 5.3.

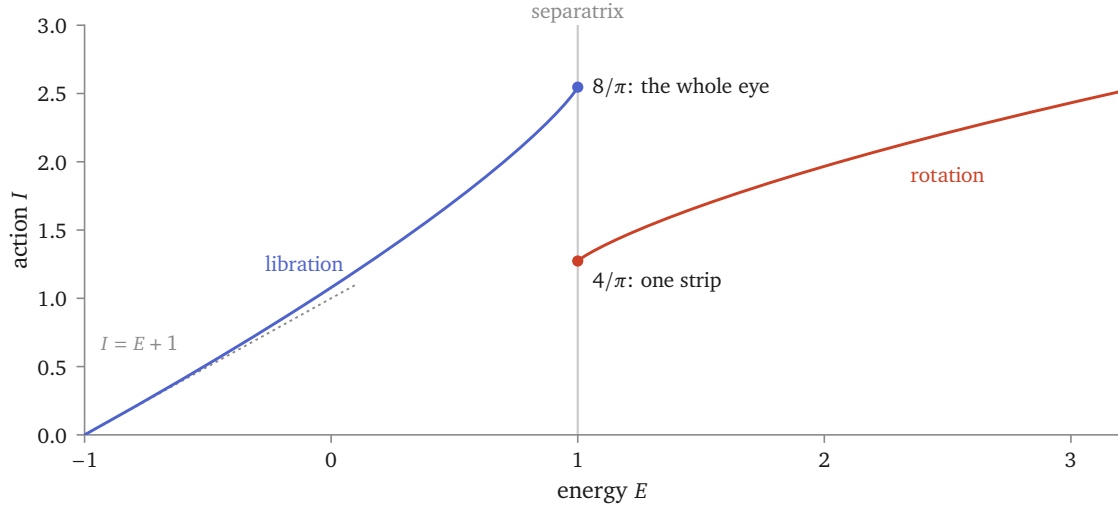


Figure 5.4: The pendulum's action $I(E)$, from (5.9) and (5.11). Near $E = -1$ it follows the oscillator's $I = E + 1$ (dotted). At the separatrix it jumps: the last libration loop encloses the whole eye, area 16; the first rotation loop sweeps one strip, area 8. The tangents are vertical there, since $dI/dE = T/2\pi$.

does not apply — this is the failure of hypothesis (I) that § 4.3 exhibited. Approaching it from either side the tori still exist, and three things happen to them.

The action jumps (Figure 5.4). As $E \uparrow 1$, $k \rightarrow 1$ and $I \rightarrow 8/\pi$, which is $1/2\pi$ times the area of the eye, $\int_{-\pi}^{\pi} 4 \cos(\theta/2) d\theta = 16$. As $E \downarrow 1$, $I \rightarrow 4/\pi$, one strip of area 8. One libration torus becomes two rotation tori, and the area is shared between them. The function I is not continuous across the separatrix because the cycle is not.

The period diverges, logarithmically (Figure 5.5). With (5.8): below, $k'^2 = (1 - E)/2$ and $T = 4K \approx 4 \ln(4/k')$; above, $k'^2 = (E - 1)/(E + 1) \approx (E - 1)/2$ and $T = 2kK \approx 2 \ln(4/k')$. So

$$T = 2 \ln \frac{32}{1 - E} + o(1) \quad (E \uparrow 1), \quad T = \ln \frac{32}{E - 1} + o(1) \quad (E \downarrow 1). \quad (5.12)$$

The coefficient on the rotation side is *half* the libration one. The reason is visible in Figure 5.3: almost all of the time is spent near the saddle, where the motion is slowest, and a libration loop passes near it twice per period — at $\theta \approx \pi$ and at $\theta \approx -\pi$, which on the cylinder are the same point — while a rotation loop passes once.

The frequency goes to zero. $\nu = 2\pi/T \rightarrow 0$: the torus degenerates into a set on which the motion takes infinitely long to go round.

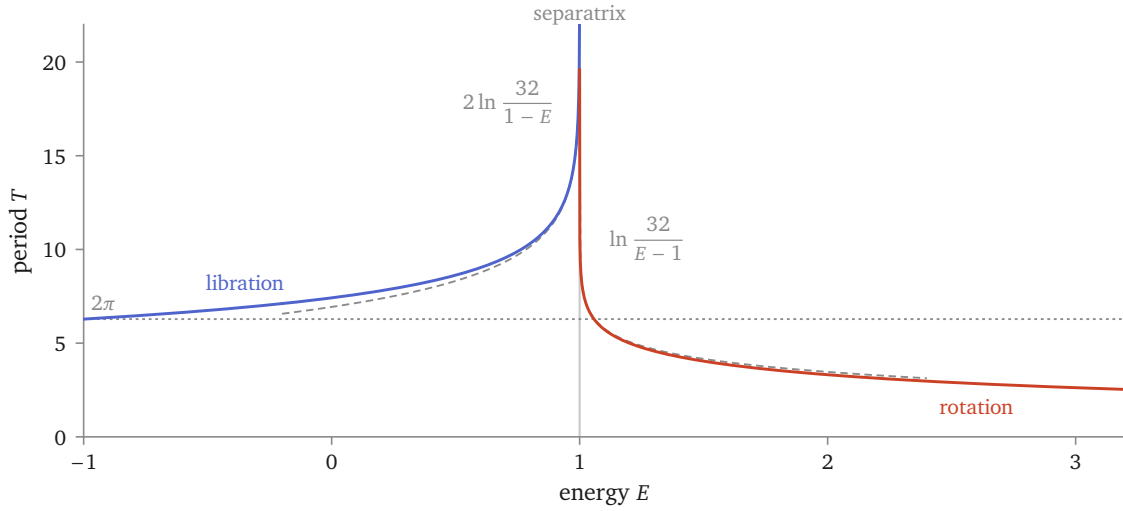


Figure 5.5: The pendulum's period $T(E)$, from (5.9) and (5.11), with the separatrix asymptotics (5.12) dashed. It starts at the oscillator's 2π (dotted) and diverges on both sides of $E = 1$, twice as fast from below as from above.

Reading

Why a logarithm. Near the saddle put $\theta = \pi + x$. Then $H \approx \frac{1}{2}p^2 - \frac{1}{2}x^2 + 1$, the inverted oscillator, whose orbits are the hyperbolas $p^2 - x^2 = 2(E - 1)$. The time to cross a fixed box $|x| \leq \delta$ along the hyperbola is $\int dx / \sqrt{x^2 + 2(E - 1)} \approx \ln(\delta^2 / |E - 1|)$, which diverges like $\ln(1/|E - 1|)$; the time spent outside the box stays bounded. One crossing per rotation, two per libration — the factor 2 in (5.12). The constant 32 needs the exact integral: it is where (5.8) is used.

5.4 Kepler

A particle in $\mathbb{R}^3 \setminus \{0\}$ with $H = \frac{1}{2}|p|^2 - \alpha/|q|$, $\alpha > 0$. In spherical coordinates (r, θ, ϕ) — θ the polar angle, not the pendulum's, and ϕ the azimuth, to be told apart from the angle variables φ —

$$H = \frac{1}{2} \left(p_r^2 + \frac{p_\theta^2}{r^2} + \frac{p_\phi^2}{r^2 \sin^2 \theta} \right) - \frac{\alpha}{r}. \quad (5.13)$$

This is a central force, integrable with $F = (H, |\ell|^2, \ell_3)$ (Example 4.7), and in these coordinates

$$\ell_3 = p_\phi, \quad |\ell|^2 = p_\theta^2 + \frac{p_\phi^2}{\sin^2 \theta}.$$

Write $\ell = |\ell|$. Fix $E < 0$ and $0 < |\ell_3| < \ell < \alpha/\sqrt{-2E}$ (the last so that the radial motion is not a circle). On the level set

$$p_\phi = \ell_3, \quad p_\theta = \pm \sqrt{\ell^2 - \frac{\ell_3^2}{\sin^2 \theta}}, \quad p_r = \pm \sqrt{2E + \frac{2\alpha}{r} - \frac{\ell^2}{r^2}}, \quad (5.14)$$

and each momentum depends only on its own coordinate: Hamilton–Jacobi separates, $S = S_r(r) + S_\theta(\theta) + \ell_3\phi$. The torus is a product of three circles — r oscillating between the

roots $r_- < r_+$ of $p_r = 0$, θ between the roots of $p_\theta = 0$, and ϕ going round — and (5.6) gives three actions (Figure 5.6):

$$I_\phi = \frac{1}{2\pi} \oint p_\phi d\phi = \ell_3, \quad I_\theta = \frac{1}{2\pi} \oint p_\theta d\theta = \ell - |\ell_3|, \quad I_r = \frac{1}{2\pi} \oint p_r dr.$$

The first is immediate. The second is a one-variable integral of the same kind as the next one; it is Sheet 5's, and we use it as stated.

Proposition 5.9 (The radial action). $I_r = \frac{\alpha}{\sqrt{-2E}} - \ell$.

Proof. $I_r = \frac{1}{\pi} \int_{r_-}^{r_+} p_r dr$, the loop being twice the passage from r_- to r_+ . Differentiate in ℓ under the integral. This is legitimate: with $r = c - h \cos s$, $c = \frac{1}{2}(r_+ + r_-)$, $h = \frac{1}{2}(r_+ - r_-)$, the limits become 0 and π and the integrand becomes $\sqrt{-2E} h^2 \sin^2 s / r$, smooth in (s, ℓ) ; transforming back gives

$$\frac{\partial I_r}{\partial \ell} = -\frac{\ell}{\pi} \int_{r_-}^{r_+} \frac{dr}{r^2 p_r} = -\frac{\ell}{\pi} \int_{u_-}^{u_+} \frac{du}{\sqrt{2E + 2\alpha u - \ell^2 u^2}}, \quad u = \frac{1}{r}.$$

The quadratic under the root is $\ell^2(u_+ - u)(u - u_-)$, and $\int_{u_-}^{u_+} du / \sqrt{(u_+ - u)(u - u_-)} = \pi$ (substitute $u = \frac{1}{2}(u_+ + u_-) - \frac{1}{2}(u_+ - u_-) \cos s$, and the integrand becomes ds). So $\partial I_r / \partial \ell = -1$ and $I_r = c(E) - \ell$. To find $c(E)$, let ℓ increase to the circular orbit, where $r_- = r_+$: the interval of integration shrinks to a point with p_r bounded, so $I_r \rightarrow 0$. The two roots of $2Er^2 + 2\alpha r - \ell^2 = 0$ coincide when $\alpha^2 + 2E\ell^2 = 0$, i.e. $\ell = \alpha / \sqrt{-2E}$. Hence $c(E) = \alpha / \sqrt{-2E}$. ■

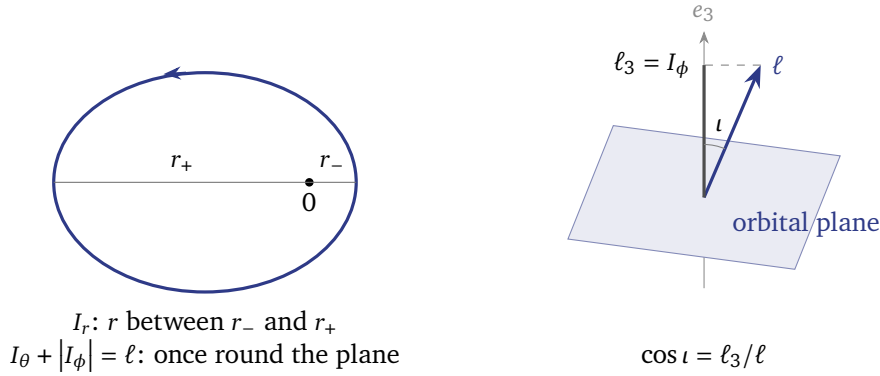


Figure 5.6: Kepler's three actions. I_r is the radial oscillation between pericentre and apocentre; $I_\theta + |I_\phi| = \ell$ is the angular momentum; $I_\phi = \ell_3$ its component along e_3 , fixing the inclination ι of the orbital plane.

Adding the three actions, $I_r + I_\theta + |I_\phi| = \alpha / \sqrt{-2E}$, so

$$K(I) = -\frac{\alpha^2}{2(I_r + I_\theta + |I_\phi|)^2}. \quad (5.15)$$

The energy depends on the actions only through their sum. By (5.3), with $\mathcal{J} = I_r + I_\theta + |I_\phi|$,

$$\nu_r = \nu_\theta = |\nu_\phi| = \frac{\alpha^2}{\mathcal{J}^3} \quad (5.16)$$

(with ν_ϕ of the sign of I_ϕ). All three frequencies are equal: every torus is resonant, in the sense of Definition 4.16, as resonant as it can be. The radial oscillation, the swing in θ and the turn in ϕ all close at the same moment, so *every bounded Kepler orbit with $\ell \neq 0$ is closed*: $\varphi(t) = \varphi(0) + \nu(1, 1, \pm 1)t$ returns to its start after $2\pi/\nu$. (The tori excluded above, $\ell_3 = 0$ or $|\ell_3| = \ell$, differ only in the choice of polar axis, and the same holds there; circular orbits are closed anyway. A bounded orbit with $\ell = 0$ is radial and falls into the centre in finite time.) The common period is $T = 2\pi\mathcal{J}^3/\alpha^2$; with the semi-major axis $a = \frac{1}{2}(r_- + r_+) = \alpha/(-2E)$ this is $T = 2\pi a^{3/2}/\sqrt{\alpha}$, Kepler's third law.

Reading

In celestial mechanics the actions are taken as $(\mathcal{J}, \ell, \ell_3)$ — the *Delaunay variables* — whose conjugate angles are the mean anomaly, the argument of pericentre and the longitude of the node. Since K depends on $\mathcal{J}$ alone, only the first angle moves. We name them and do not use them.

What a perturbation breaks

Add a small β/r^2 to the potential. It only replaces ℓ^2 by $\ell^2 + 2\beta$ in p_r , so the proof of Proposition 5.9 goes through unchanged and

$$I_r = \frac{\alpha}{\sqrt{-2E}} - \sqrt{\ell^2 + 2\beta}, \quad K = -\frac{\alpha^2}{2(I_r + \sqrt{\ell^2 + 2\beta})^2}.$$

The motion is still in a plane, and ℓ is the action of the angle in it. Now $\partial K/\partial \ell = \nu_r \ell/\sqrt{\ell^2 + 2\beta}$: the angular frequency differs from the radial one. In one radial period the orbit turns by $2\pi\ell/\sqrt{\ell^2 + 2\beta}$, not 2π , so the pericentre moves on every pass, and unless the ratio is rational the orbit never closes; it fills the annulus $r_- \leq r \leq r_+$ densely (Arnold, §8 C) (Figure 5.7). The degeneracy (5.16) is the whole reason Kepler orbits close, and it is fragile.

5.5 Superintegrability and Laplace–Runge–Lenz

Equal frequencies are the footprint of *extra* integrals. For Kepler the extra one is the *Laplace–Runge–Lenz vector*

$$A = p \times \ell - \alpha \frac{q}{|q|}. \quad (5.17)$$

It is conserved, $\{A_i, H\} = 0$ (Sheet 5); it lies in the orbital plane, $\ell \cdot A = 0$; and it points from the centre to the pericentre, with length $|A| = \alpha e$ (Figure 5.8). A conserved vector pointing at the pericentre is exactly the statement that the pericentre does not move.

Count the integrals. H , the three ℓ_i and the three A_i are seven functions on a six-dimensional phase space, tied by two relations,

$$\ell \cdot A = 0, \quad |A|^2 = \alpha^2 + 2H |\ell|^2,$$

and 5 of them are independent. The Jacobian has rank 5 at one sample point,² so some 5×5 minor is not identically zero; the minors are real-analytic on the connected set $q \neq 0$, and such a function vanishing on an open set vanishes identically, so the minor is nonzero on a dense open set.

²`scripts/check-l5-algebra.py`, check 7: the relations symbolically, the rank exactly at a rational sample point.

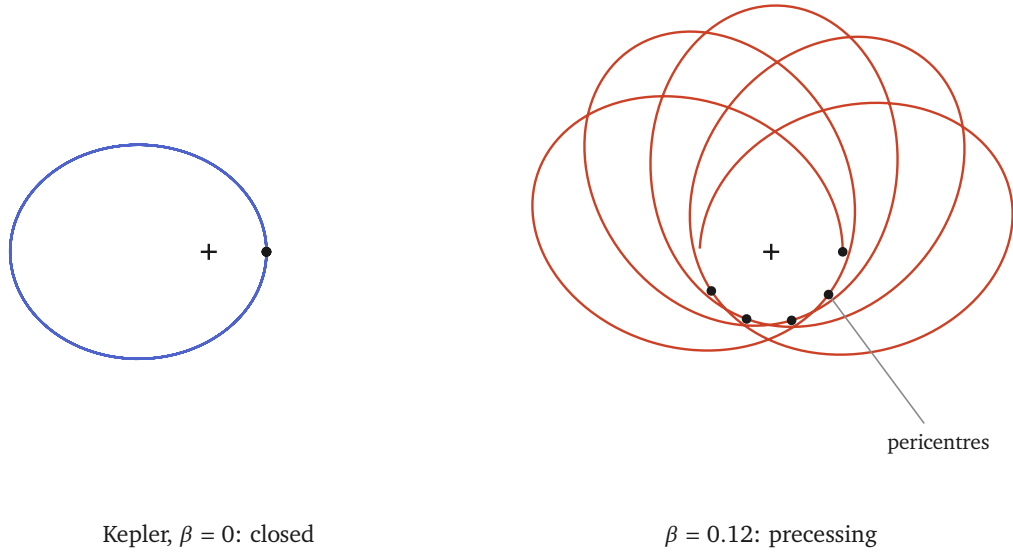


Figure 5.7: Five radial periods, with the same α , ℓ and e . Left, Kepler: the radial and angular frequencies are equal and the ellipse retraces itself. Right, with β/r^2 added ($\beta = 0.12$, $\ell = 1$): the angle between successive pericentres (dots) is $2\pi\ell/\sqrt{\ell^2 + 2\beta} < 2\pi$. Both curves are exact orbits (`scripts/figures/kepler_precession.py`).

Definition 5.10 (Superintegrable). An integrable system with n degrees of freedom is *superintegrable* if it has more than n independent first integrals, and *maximally superintegrable* if it has $2n - 1$.

Kepler is maximally superintegrable: $5 = 2 \cdot 3 - 1$. The oscillator with equal frequencies is the other standard example.

Reading

The extra integrals explain the closed orbits a second way. Fix $E < 0$ and $\ell \neq 0$. Then ℓ fixes the orbital plane and the sense of motion, A the eccentricity $e = |A|/\alpha$ and, if $e > 0$, the direction of the pericentre, and H the size: one ellipse in q -space. Over each of its points p is fixed as well, by $p = \ell \times (A + \alpha q/|q|)/|\ell|^2$, which inverts (5.17) because $p \perp \ell$. So the joint level set of (H, ℓ, A) is a single closed curve in phase space. X_H does not vanish on it ($p \neq 0$ since $\ell \neq 0$), so the orbit is the whole curve and is periodic. The integrals do not all commute: $\{\ell_i, A_j\} = \sum_k \epsilon_{ijk} A_k$ and $\{A_i, A_j\} = -2H \sum_k \epsilon_{ijk} \ell_k$ (Sheet 5), so at most $n = 3$ of them can be the integrals of Theorem 4.8 at once.

The other side of the coin is the frequency map. Call an integrable system *non-degenerate* if $I \mapsto \nu(I)$ is a local diffeomorphism, $\det(\partial^2 K/\partial I^2) \neq 0$. For $n \geq 2$ the resonant tori, the preimages of the rational directions, are then dense but form a set of measure zero (Figure 5.9); for $n = 1$ resonance means $\nu = 0$, and there may be none at all. Lecture 6 needs exactly this picture. Kepler is as degenerate as possible: its frequency map has rank one, and every torus is resonant.

Which central potentials share Kepler's luck? *Bertrand's theorem*: among smooth central potentials that have bounded orbits at all, the only ones all of whose bounded non-collision orbits are closed are, up to an additive constant, $-\alpha/r$ with $\alpha > 0$ and $\frac{1}{2}\nu^2 r^2$ (cited: Arnold, §8 D, through a sequence of problems).

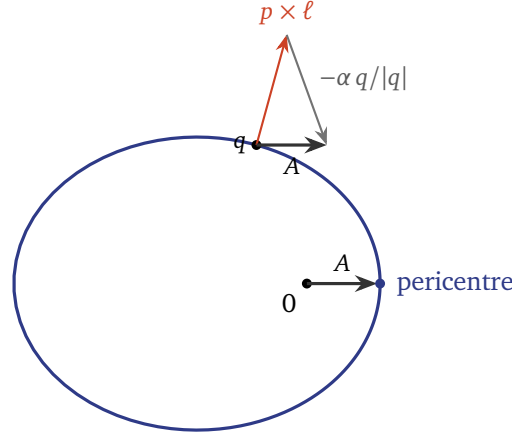


Figure 5.8: The Laplace–Runge–Lenz vector on an orbit with $\alpha = \ell = 1$, $e = 0.6$. At q the two terms of (5.17), drawn to scale, add to the same A as at the pericentre.

5.6 Bridge to Part II: Bohr–Sommerfeld

The old quantum theory quantised the actions,

$$\oint_{\gamma_i} p \cdot dq = 2\pi\hbar n_i, \quad n_i \in \mathbb{Z}, \quad \text{that is,} \quad I_i = \hbar n_i. \quad (5.18)$$

For the oscillator it gives $E = \hbar\nu n$, which misses the zero-point energy $\frac{1}{2}\hbar\nu$; for Kepler, (5.15) with $\mathcal{J} = \hbar n$, $n = 1, 2, \dots$, gives $E = -\alpha^2/2\hbar^2 n^2$, the Bohr levels. The correction $n \mapsto n + \frac{1}{2}$ at each turning-point cycle, and the reason (5.18) works as well as it does, belong to Part II. What matters now is its shape: *quantisation is a condition on the actions*, and [Lecture 15](#) will find the Bethe equations to be this condition for a many-body system.

Summary.

- Actions $I_i = \frac{1}{2\pi} \oint_{\gamma_i} p \cdot dq$ ([Definition 5.1](#)) are well defined because $p \cdot dq$ is closed on the torus, and are functions of F in involution ([Proposition 5.2](#)).
- The period formula $\partial I / \partial f = A^\top / 2\pi$ ([Proposition 5.3](#)) makes the actions independent, so $H = K(I)$, and the frequencies are $\nu = \partial K / \partial I$ ([Corollary 5.4](#)); for one degree of freedom, $dI/dE = T/2\pi$.
- Angles are $\varphi = \partial S / \partial I$ for the generating function $S(q, I) = \int p \cdot dq$ along the torus, which solves Hamilton–Jacobi ([Theorem 5.5](#)). When Hamilton–Jacobi separates, the actions are one-variable integrals (5.6).
- Oscillator: $I = E/\nu$, frequency independent of the action. Pendulum: I and T in terms of $K(k)$ and $E(k)$, in two regimes with different cycles ([Propositions 5.7](#) and [5.8](#)); at the separatrix I jumps from $8/\pi$ to $4/\pi$ and T diverges like $2 \ln(32/(1-E))$ below and $\ln(32/(E-1))$ above (5.12).
- Kepler: $I_r = \alpha/\sqrt{-2E} - \ell$ ([Proposition 5.9](#)), $K = -\alpha^2/2(I_r + I_\theta + |I_\phi|)^2$, all frequencies equal, every bounded orbit closed. A β/r^2 perturbation splits the frequencies and the orbit precesses.
- Superintegrability: the Laplace–Runge–Lenz vector gives Kepler $5 = 2n - 1$ independent integrals ([Definition 5.10](#)); Bertrand’s theorem (cited). Bohr–Sommerfeld $I = \hbar n$ (5.18) is the bridge to [Lecture 15](#).

Tutorial. Exercise Sheet 5.

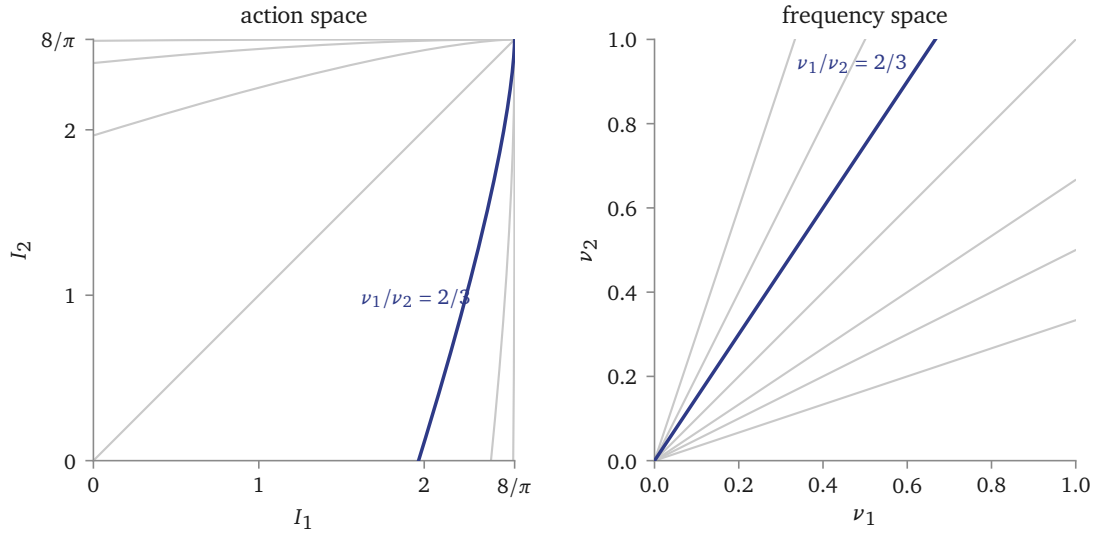


Figure 5.9: The frequency map of two uncoupled librating pendula, $K = K_{\text{pend}}(I_1) + K_{\text{pend}}(I_2)$, computed from (5.9). It is non-degenerate: $\nu(I)$ falls strictly from 1 at $I = 0$ to 0 at the separatrix $I = 8/\pi$. The resonant tori $\nu_1/\nu_2 = a/b$ are lines through 0 in frequency space and curves in action space; one is marked.

References

Arnold, *Mathematical Methods of Classical Mechanics*, §50 (action–angle variables: the construction and the neighbourhood statement) and §8 (central fields: §8 C the dense annulus, §8 D Bertrand’s theorem, through a sequence of problems); Babelon, Bernard and Talon, *Introduction to Classical Integrable Systems*, ch. 2; Dunajski, *Integrable Systems*, §1.2 (action–angle variables, and Kepler’s radial action by contour integration — a different route to Proposition 5.9); DLMF, §§19.2 and 19.12 (complete elliptic integrals and their behaviour as $k \rightarrow 1$). **Conventions.** Arnold writes $\omega = \sum dp \wedge dq$, the negative of ours, and so $\omega = d(p dq)$ where we have $-d(p \cdot dq)$; his actions come out positive for the same reason ours do (§ 3.7).